Business Intelligence Overview Report

Superstore Executive Sales Dashboard | Prepared by Divine Okezie Nwaoha | AnalystLab Africa Experience Lab, Week 2

1. What Is Business Intelligence?

Business Intelligence (BI) is the practice of collecting, cleaning, organising and presenting data so that people can understand how a business is performing and decide what to do next. It turns raw records, such as thousands of individual sales transactions, into clear measures like total sales, profit margin and regional performance. A typical BI process has four steps: gather the data, prepare it (remove duplicates, fix data types, handle missing values), analyse it, and present the results in a form that decision-makers can use.

2. Why Organisations Use Dashboards

A dashboard is a single screen that brings the most important measures together, usually as key performance indicators (KPIs), charts and filters. Organisations use dashboards for four main reasons:

- **Speed:** managers see the current position at a glance instead of reading long spreadsheets or reports.
- **One version of the truth:** everyone works from the same cleaned data and the same definitions, such as how profit margin is calculated.
- **Interactivity:** slicers let a user filter by year, region, category or segment and see the answer immediately, without asking an analyst for a new report.
- **Early warning:** problems such as a loss-making product or region show up visually (for example, in red) before they grow into larger losses.

3. How Dashboards Improve Decision-Making

Dashboards improve decisions because they replace opinion with evidence and make comparison easy. A manager can compare regions, categories and years side by side, see which areas are growing or shrinking, and judge where to spend time and money. They also shorten the gap between a problem appearing and a decision being made, because the information is already organised and visual. In this project, for example, the dashboard

shows that Furniture earns only a 2.5% profit margin despite $0.74M in sales, and that five states lose money. Those findings point management directly to pricing, discount and regional reviews, which would be hard to spot in raw transaction data.

4. Business Objective of the Selected Dataset

The project uses the Superstore Sales dataset (Option 1), which holds the order-level records of a national US retail company from 2014 to 2017: about 9,993 order lines covering sales, profit, discounts, quantities, customers, products, regions and states.

The business objective is to give senior management an interactive executive dashboard that answers seven questions:

No.	Business question	Where the dashboard answers it
1	What is the overall sales performance?	KPI cards: Total Sales, Total Profit, Total Orders, Average Sales, Profit Margin (Sales Overview)
2	Which regions generate the highest sales and profit?	Sales by Region bar chart, profit margin matrix, Profit by State map
3	Which customer segments contribute the most revenue?	Total Sales by Segment column chart, Segment slicer
4	Which product categories perform best?	Total Sales by Category column chart, profit margin matrix
5	Which products are the most profitable?	Top 10 Sub-Categories by Profit bar chart
6	What trends appear over time?	Monthly Sales Trend by Year line chart, Year slicer
7	What should management do to improve performance?	Executive Insights page: insights, risks, opportunities, recommendations

The finished dashboard has three pages that together support the central goal: growing sales while protecting profit.

Page	Purpose	Main visuals
Sales Overview	Headline performance and filters	5 KPI cards, 5 slicers, Sales by Category and Segment columns, Sales by Region bar chart
Trends and Profitability	Trends and profit drivers	Monthly sales line chart, Ship Mode donut, Top 10 Sub-Categories bar chart, Profit by State map
Executive Insights	Findings and actions	Profit margin matrix, key insights, risks, opportunities, recommendations

5. Tools and Method

The data was imported into Microsoft Power BI, inspected in Power Query, and cleaned by correcting data types (dates, decimals, text), checking for blank values and removing one duplicate record. DAX measures were created for Total Sales, Total Profit, Total Orders, Average Sales and Profit Margin. The dashboard was then built with column, bar, line and donut charts, a map, a matrix, and five linked slicers.